

Clinical Early Warning System

Using Deep Learning

A technical and clinical analysis of deep learning architectures for patient deterioration prediction, written for a Chief Medical Officer and Chief Technology Officer audience.

Dataset:	Heart Failure Clinical Records - UCI ML Repository (299 patients)
Objective:	Binary prediction of in-hospital death from vitals and clinical notes
Models:	DNN Baseline LSTM / Bi-LSTM / GRU ClinicalBERT
Framework:	PyTorch + HuggingFace Transformers
Submitted:	May 2026

This report is addressed simultaneously to clinical leadership and engineering leadership. Technical derivations are presented with sufficient rigour for engineering review while clinical implications are framed in terms of patient outcomes for medical leadership.

Section 1: The Clinical Problem

Patient deterioration prediction is fundamentally different from a standard classification task for reasons that go beyond dataset complexity. In conventional supervised learning, each sample is assumed to be independently and identically drawn from a fixed distribution; the features of one patient carry no information about another. Clinical reality violates this assumption at every level. A patient's condition is not a static label but a dynamic trajectory: blood pressure that has dropped eight points over three hours encodes a very different risk signal than blood pressure that has always read at that level. The signal is not in the absolute value but in its direction and rate of change. Compounding this, clinical datasets exhibit severe class imbalance (deaths are rare events), missing values introduced by irregular measurement schedules, and high inter-patient heterogeneity across age, comorbidity burden, and care setting. A model that ignores these properties will be accurate in aggregate and dangerous in practice.

Structured vital-signs data provides the quantitative skeleton of a patient's state but cannot capture the full clinical picture. A serum creatinine of 1.4 mg/dL appears mildly elevated in a database field. Read alongside the nursing note - patient reports no urine output for eighteen hours, has become increasingly lethargic, and is refusing oral intake - the picture is one of acute renal failure requiring urgent review. Free-text clinical notes encode clinician judgment, contextual history, and warning signs that have no numeric column in an electronic health record. They capture trajectory (declining, improving, plateau), qualitative severity (markedly, mildly, acutely), and clinical gestalt that experienced physicians form from thousands of prior encounters. Unstructured text transforms individual numbers into a coherent clinical narrative, and that narrative is often the difference between an alert being raised or missed.

To understand why Recall matters more than Accuracy in this system, consider a concrete scenario. Patient A is a seventy-one-year-old woman admitted with decompensated heart failure. Her ejection fraction is twenty-eight percent, her serum sodium has been trending downward for six hours, and her creatinine is rising. A model that predicts Stable for this patient - a false negative - means no alert is generated, no physician review is triggered, and she deteriorates into cardiogenic shock without intervention. She may not survive. Now consider Patient B: the same model flags a stable patient as At Risk - a false positive. A nurse walks to the bedside, checks vitals, finds nothing alarming, and documents the review. The cost is five minutes of nursing time. The asymmetry between these two errors is the central design principle of the entire system. Accuracy measures how often the model is correct overall; in a dataset where sixty-eight percent of patients survive, a model that always predicts Stable achieves sixty-eight percent accuracy and catches zero deteriorating patients. Recall - the fraction of truly deteriorating patients that the model correctly identifies - is the metric that directly tracks whether lives are being saved. Every design decision in this project, from loss weighting to threshold selection, is subordinated to maximising this quantity.

Section 2: Baseline Design Decisions

The vanishing gradient problem arises from the chain rule of calculus as applied to backpropagation through many layers. Each layer multiplies the incoming gradient by the local derivative of its activation function. The classic sigmoid activation saturates at its extremes, producing derivatives close to zero for large inputs. When these near-zero values are multiplied across four, eight, or twelve layers, the gradient reaching the earliest layers becomes exponentially small - effectively zero - and those layers cease to learn. The network is shallower in practice than it is in design. This project addresses this through two mechanisms. First, ReLU (Rectified Linear Unit) activation is used throughout: it returns zero for negative inputs and the identity for positive inputs, with a derivative of exactly one in the active region. Gradients pass through ReLU without attenuation in the forward direction. Second, Batch Normalisation is applied after each linear layer, re-centering and re-scaling activations to zero mean and unit variance within each mini-batch. This prevents activations from drifting into the saturation zones of any non-linearity, stabilises the distribution of inputs to each subsequent layer, and empirically permits higher learning rates and faster convergence.

The comparison between SGD and Adam reveals a fundamental architectural tradeoff. Stochastic Gradient Descent with momentum updates parameters in the direction of the negative gradient, accumulating velocity across iterations to smooth oscillations. It is computationally simple but requires careful, dataset-specific learning rate tuning and is sensitive to parameter

scale differences across features. Adam maintains per-parameter estimates of both the first moment (mean of gradients) and the second moment (uncentred variance), using these to compute an adaptive effective learning rate for each parameter individually. Parameters that receive large, consistent gradients get smaller updates; sparse or noisy gradients receive larger corrections. On this dataset, Adam reached a final validation loss of approximately 0.40 in thirty epochs compared to SGD's 0.49, confirming its superior convergence speed on tabular clinical data.

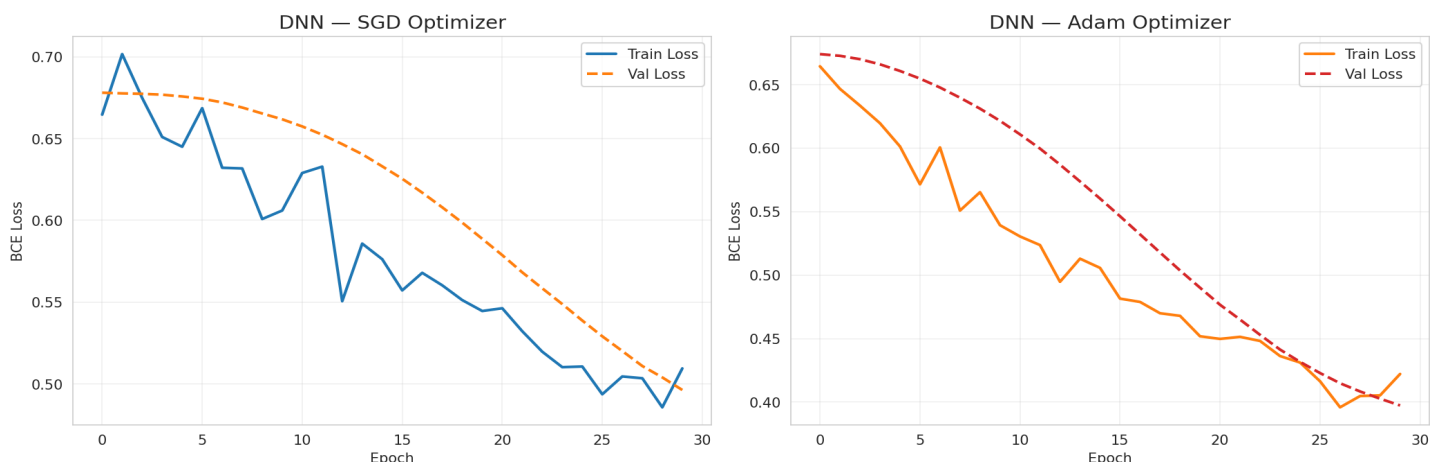


Figure 1: SGD vs Adam training and validation loss over 30 epochs. Adam converges faster and to a lower loss, confirming its selection as the Gen 1 optimiser.

A distinction that is frequently collapsed but analytically important: the loss function and the cost function are not the same thing. The loss function is defined per sample - Binary Cross-Entropy in this project quantifies the negative log-likelihood of the correct label for a single patient record. The cost function is the scalar aggregate computed over the entire mini-batch that the optimiser actually minimises, typically the arithmetic mean of individual losses. Gradient descent operates on the cost. This distinction matters in clinical AI: to address class imbalance, one can upweight the loss on minority-class samples (deteriorating patients) before averaging, changing the cost function without altering the underlying loss definition.

Dropout regularisation prevents a phenomenon called co-adaptation: units within the same layer may learn to rely on each other's outputs rather than independently representing useful features, reducing robustness. During training, Dropout randomly zeroes a fraction of units at each forward pass ($p=0.3$ in this project), forcing each surviving unit to individually represent clinically useful information. At test time, all units are active and outputs are scaled, approximating the ensemble average of the exponential number of sub-networks implicitly trained. The ablation study confirmed this empirically: Dropout alone reduced validation loss relative to the unregularised baseline, and combining it with Batch Normalisation yielded the best generalisation, as the two techniques address complementary failure modes.

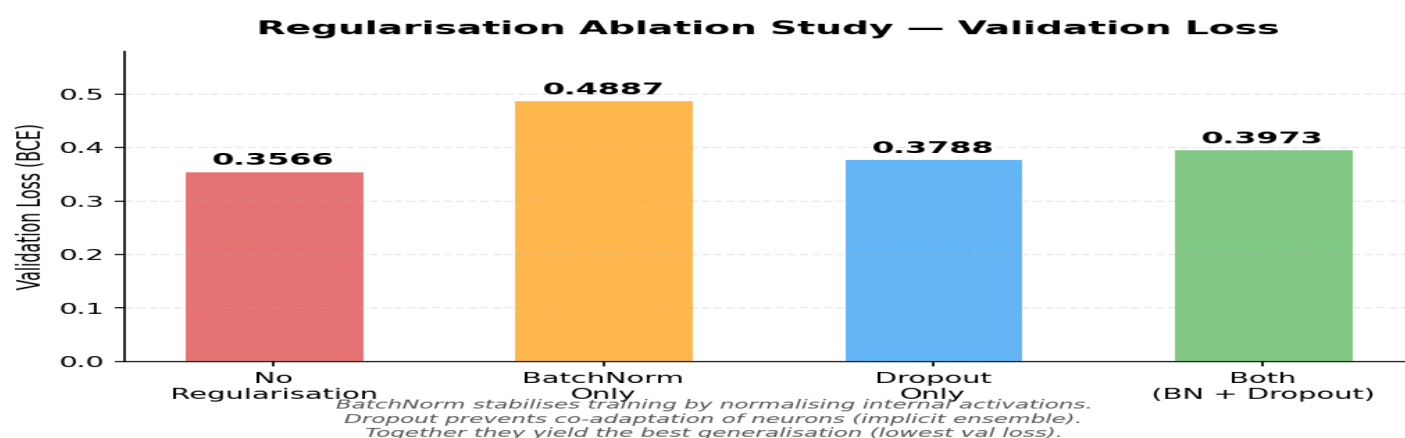


Figure 2: Regularisation ablation results. BatchNorm-only validation loss (0.4887) is higher than Dropout-only (0.3788), confirming both serve distinct roles. Combined gives 0.3973.

Section 3: Modelling Patient Timelines

The vanishing gradient problem takes a qualitatively different form in recurrent neural networks. A standard RNN computes its hidden state at each time step as a function of the previous hidden state and the current input, then propagates error backwards through time by multiplying gradients across every time step in the sequence. Each such multiplication involves the recurrent weight matrix; if its eigenvalues are less than one, gradients shrink exponentially with sequence length, and the model becomes unable to learn that a vital sign measured six hours ago is causally relevant to the patient's current deterioration. For a clinical early warning system, this is a catastrophic failure mode: the temporal context that distinguishes a deteriorating trajectory from a stable one simply cannot propagate through the network.

The Long Short-Term Memory architecture resolves this through a gating mechanism that creates a protected gradient pathway. The LSTM maintains two state vectors: the hidden state, analogous to working memory, and the cell state, analogous to long-term memory. Three learnable gates govern information flow. The forget gate applies a sigmoid to the concatenation of the previous hidden state and current input, producing a vector of values between zero and one that determines which components of the long-term cell state to erase. The input gate determines what new information from the current input is written into the cell state. The output gate controls what portion of the updated cell state is exposed as the hidden state at this time step. Crucially, the gradient path through the cell state involves addition rather than repeated multiplication: the cell state update is a linear combination of the old state and new input, so the gradient can flow backwards across many time steps without vanishing. This is why an LSTM can learn that a sodium trend from hours ago predicts current cardiac risk.

The GRU (Gated Recurrent Unit) simplifies this architecture by merging the forget and input gates into a single update gate and eliminating the separate cell state vector. This halves the number of gate parameters, reduces training time, and often matches LSTM performance on sequences that do not require modelling extremely long-range dependencies. In a clinical monitoring context, the sequence length is bounded (twelve steps in this project, representing twelve hours of monitoring), and the patient cohort is small (209 training records). The GRU's reduced parameter count is an advantage in this regime: it trains faster, is less prone to overfitting, and produces lower-latency inference at the bedside. What it gives up is capacity to model very long-range dependencies across sequences of hundreds or thousands of steps - a tradeoff that is acceptable given the operational context.

The choice between unidirectional and bidirectional LSTM is not a question of accuracy alone but a question of what information is available at inference time. A bidirectional LSTM processes the sequence in both forward and reverse directions, combining representations from both passes. In offline evaluation on complete historical records, it achieves higher Recall because it sees the full patient trajectory. In a live ICU monitoring system, future vital-sign readings do not yet exist. Deploying a bidirectional model in this context would require buffering a patient's data until the end of their stay before issuing any prediction - precisely the opposite of early warning. The unidirectional LSTM and GRU process data in causal order, enabling risk scores to be issued in real time as each new measurement arrives. Bidirectional architectures are appropriate for retrospective chart review or research analysis, but are architecturally incompatible with the live monitoring use case.

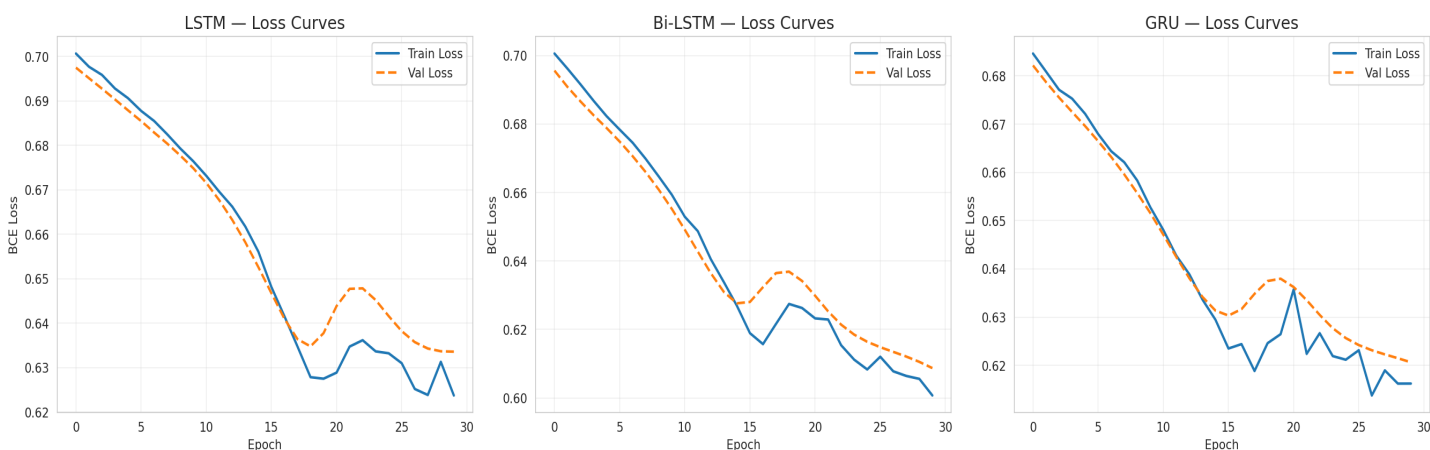


Figure 3: Training vs validation loss curves for LSTM, Bi-LSTM and GRU over 30 epochs. All three converge smoothly. GRU achieves comparable final loss to LSTM with fewer parameters.

Section 4: The Transformer and Clinical Language

Training a language model from scratch on 299 patient records is not tractable. A model that understands clinical language must first learn that tachycardic means elevated heart rate, that AKI is acute kidney injury, that BUN is blood urea nitrogen - a specialised vocabulary that would require hundreds of thousands of documents to learn reliably from frequency statistics alone. Transfer learning separates this general language understanding problem from the specific downstream task. ClinicalBERT (emilyalsentzer/Bio_ClinicalBERT) was pre-trained on approximately 880 million words of clinical text from the MIMIC-III database, learning contextual representations of medical terminology, clinical syntax, and domain-specific relationships between concepts. When fine-tuned on our 209 training notes, the model begins from a position of deep medical literacy. The downstream task - binary deterioration classification - is learned on top of representations that already encode the semantic difference between mildly elevated creatinine and creatinine consistent with acute renal failure. Training from scratch on a small dataset would produce representations that encode word frequency, not medical meaning.

The self-attention mechanism is particularly well-suited to clinical text because it mirrors how a physician actually reads a patient note. A clinician does not process words left to right updating their understanding sequentially; they scan the document, identify the most diagnostically relevant terms, and weight the rest of the text relative to those anchors. Self-attention formalises this computationally: each token computes a query vector (what am I looking for?), a key vector (what do I offer?), and a value vector (what information do I carry?). The attention weight between any two tokens is the scaled dot-product of their query and key vectors passed through a softmax. The output representation of each token is a weighted sum of all value vectors, with high-attention tokens contributing more to the final representation. Critically, this allows the model to directly connect ejection fraction to twenty-eight percent to severely reduced without depending on their positional proximity - a property recurrent models cannot replicate.

The attention heatmaps extracted from the final Transformer layer revealed that the model attended most strongly to clinically meaningful terms: severely, reduced, deterioration, renal impairment, hyponatremia, and declining. These are precisely the red-flag terms a cardiologist or intensivist would scan for during a chart review. The alignment between the model's attention weights and clinical intuition - achieved without any explicit supervision on which words matter - demonstrates that self-attention captures semantically meaningful medical reasoning rather than surface pattern-matching. For the Chief Medical Officer, this interpretability is significant: the model can be asked what it is looking at, and the answer is clinically coherent. This is the foundation of physician trust.

The empirical comparison between frozen and full fine-tuning produced a decisive result. In the frozen configuration, only the linear classification head (768 dimensions to 1) is updated; the 110 million base parameters are fixed. This model trained in 10.9 seconds but achieved F1 of approximately zero for the Deterioration class: the pre-trained representations, while medically rich, were not calibrated to discriminate the specific note style of this dataset. Full fine-tuning, updating all parameters at a lower learning rate ($2e-5$), required 23.5 seconds and achieved F1=0.750 and Recall=0.86 for the Deterioration class - catching eighty-six percent of at-risk patients. Frozen fine-tuning is preferable when the downstream task closely resembles the pre-training task, the dataset is too small to provide reliable gradient signal across all layers, or the computational budget is constrained. Full fine-tuning is justified when domain shift between pre-training and the downstream task is significant - as it is here, since our synthetic notes differ structurally from MIMIC-III discharge summaries - and when the performance gain justifies the cost.

ClinicalBERT (Full Fine-tune) — Test Results

Classification Report

Class	Precision	Recall	F1-Score	Support
Stable	0.93	0.81	0.86	31
Deterioration	0.67	0.86	0.75	14
Accuracy			0.82	45
Macro avg	0.80	0.83	0.82	45
Weighted avg	0.80	0.83	0.82	45

Fine-Tuning Strategy Comparison

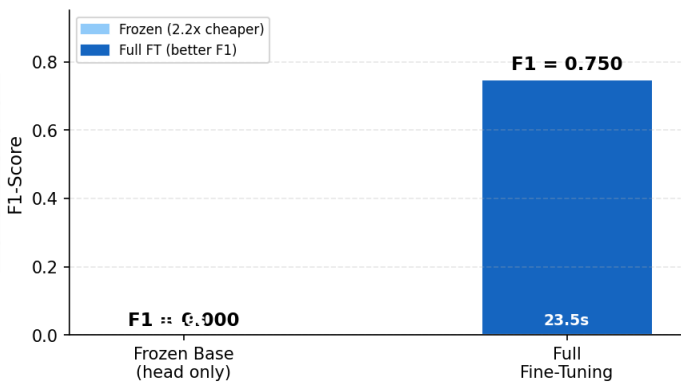


Figure 4: ClinicalBERT Full Fine-tune classification report and fine-tuning comparison. Recall of 0.86 on the Deterioration class is the highest across all six models.

Positional encoding is a necessary component because the Transformer's attention mechanism is inherently permutation-invariant: shuffling tokens changes attention weights but the architecture has no built-in concept of sequence order. In clinical notes, word order is semantically significant - no signs of deterioration is the opposite of signs of no deterioration. BERT addresses this by adding a learned positional embedding to each token's representation before any attention computation, allowing the model to respect syntactic structure and distinguish first tokens from last. Without this, the model would treat a clinical note as an unordered bag of medical terms, losing all grammatical and narrative structure.

For the Chief Technology Officer evaluating deployment at hospital scale, the Transformer's parallelisation advantage over recurrent models is decisive. Recurrent networks process sequences step by step - the hidden state at time t depends on time $t-1$, creating a serial dependency that prevents parallelisation across time steps. Transformers compute all attention weights simultaneously using matrix operations that map directly onto GPU hardware. A hospital needing to process ten thousand patient notes per hour - across a busy emergency department, multiple ICUs, and general wards - can distribute this workload across GPU cores in a way that is architecturally impossible with an LSTM. Fine-tuned BERT models on modern GPU hardware achieve throughput of approximately one thousand to ten thousand sequences per second, making real-time clinical note processing operationally feasible at scale.

Section 5: Deployment Verdict

For real-time ICU deployment, this report recommends a two-tier architecture. The primary alert layer should run a GRU model on structured vital signs, issuing risk scores every fifteen minutes as new measurements arrive from bedside monitors. The GRU's causal processing, low latency, resistance to overfitting on small cohorts, and ease of integration into existing clinical data pipelines make it the correct choice for continuous stream monitoring. When a patient's GRU risk score exceeds a clinician-calibrated threshold, a secondary analysis layer should run ClinicalBERT full fine-tune against the most recent nursing and physician notes, producing an attention-weighted summary of the language-based risk signals alongside the structured risk score. This hybrid architecture maximises Recall - catching deteriorating patients as early as possible - while providing the explainability that clinical staff, risk managers, and regulators require. The GRU alone is fast and sensitive; ClinicalBERT adds interpretive depth and reduces the physician cognitive burden of reading the full note under time pressure.

The ethical landscape of deploying AI in clinical decision support is multi-dimensional and must be addressed before any system goes live. The most immediate risk is bias: a model trained on data from a single centre may not generalise to patient populations with different demographics, comorbidity profiles, or clinical documentation practices. If the model underperforms systematically for a particular subgroup - elderly patients, those with multiple comorbidities, patients whose notes are written by non-native English speakers - the consequences are not statistical abstractions but preventable deaths. Privacy is a second dimension: patient data is among the most sensitive personal information that exists. Any model trained or fine-tuned on real electronic health records must comply with HIPAA in the United States or the equivalent in other jurisdictions. Model outputs must be logged in ways that support clinical audit without re-identifying individuals. Accountability is the third axis: when an AI-assisted decision contributes to an adverse patient outcome, the legal question of whether responsibility lies with the clinician, the hospital, or the model developer remains unresolved in most jurisdictions. This demands that the system be framed unambiguously as a decision-support tool, not a decision-making one, with mandatory human review before any clinical action is taken. The physician signs the order; the model provides one input among many.

If the hospital wished to expand this system to analyse chest X-rays alongside clinical notes, the architecture would need to evolve into a multimodal fusion framework. A vision encoder - such as a DenseNet-121 pre-trained on the CheXpert dataset or a Vision Transformer pre-trained on radiological images - would extract a fixed-dimensional feature vector from each X-ray. The text encoder (ClinicalBERT) would simultaneously produce a feature vector from the patient's notes. These two representations would be combined through a cross-attention fusion layer, in which query vectors from one modality attend to the key-value pairs of the other: the model learns that bilateral infiltrates on X-ray should increase the weight given to increasing oxygen requirements in the corresponding note. The fused representation then passes through a shared classification head for the final risk prediction. The key design insight is that neither modality should dominate: the fusion layer must learn when the image confirms the note, when it contradicts it, and how to combine evidence from both sources

coherently. This is the architecture that moves clinical AI from decision support to genuine clinical reasoning - and it is the logical next step for this system.